

BP Statistical Review of World Energy

BP Statistical Review of World Energy June 2016

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Approximate conversion factors

Crude oil*

From	To				
	tonnes (metric)	kilolitres	barrels	US gallons	tonnes per year
	Multiply by				
Tonnes (metric)	1	1.165	7.33	307.86	–
Kilolitres	0.8581	1	6.2898	264.17	–
Barrels	0.1364	0.159	1	42	–
US gallons	0.00325	0.0038	0.0238	1	–
Barrels per day	–	–	–	–	49.8

*Based on worldwide average gravity.

Products

	To convert			
	barrels to tonnes	tonnes to barrels	kilolitres to tonnes	tonnes to kilolitres
	Multiply by			
Liquefied petroleum gas (LPG)	0.086	11.60	0.542	1.844
Gasoline	0.120	8.35	0.753	1.328
Kerosene	0.127	7.88	0.798	1.253
Gas oil/diesel	0.134	7.46	0.843	1.186
Residual fuel oil	0.157	6.35	0.991	1.010
Product basket	0.125	7.98	0.788	1.269

Natural gas (NG) and liquefied natural gas (LNG)

From	To					
	billion cubic metres NG	billion cubic feet NG	million tonnes oil equivalent	million tonnes LNG	trillion British thermal units	million barrels oil equivalent
	Multiply by					
1 billion cubic metres NG	1	35.3	0.90	0.74	35.7	6.60
1 billion cubic feet NG	0.028	1	0.025	0.021	1.01	0.19
1 million tonnes oil equivalent	1.11	39.2	1	0.82	39.7	7.33
1 million tonnes LNG	1.36	48.0	1.22	1	48.6	8.97
1 trillion British thermal units	0.028	0.99	0.025	0.021	1	0.18
1 million barrels oil equivalent	0.15	5.35	0.14	0.11	5.41	1

Units

1 metric tonne	= 2204.62lb = 1.1023 short tons
1 kilolitre	= 6.2898 barrels = 1 cubic metre
1 kilocalorie (kcal)	= 4.187kJ = 3.968Btu
1 kilojoule (kJ)	= 0.239kcal = 0.948Btu
1 British thermal unit (Btu)	= 0.252kcal = 1.055kJ
1 kilowatt-hour (kWh)	= 860kcal = 3600kJ = 3412Btu

Calorific equivalents

One tonne of oil equivalent equals approximately:

Heat units	10 million kilocalories 42 gigajoules 40 million British thermal units
Solid fuels	1.5 tonnes of hard coal 3 tonnes of lignite
Gaseous fuels	See Natural gas and liquefied natural gas table
Electricity	12 megawatt-hours

One million tonnes of oil or oil equivalent produces about 4400 gigawatt-hours (= 4.4 terawatt-hours) of electricity in a modern power station.

1 barrel of ethanol = 0.57 barrel of oil
1 barrel of biodiesel = 0.88 barrel of oil

OECD members

Europe: Austria, Belgium, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Luxembourg, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey, UK.

Other member countries: Australia, Canada, Chile, Israel, Japan, Mexico, New Zealand, South Korea, US.

OPEC members

Middle East: Iran, Iraq, Kuwait, Qatar, Saudi Arabia, United Arab Emirates.

North Africa: Algeria, Libya.

West Africa: Angola, Nigeria.

South America: Ecuador, Venezuela.

European Union members

Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, UK.

Non-OECD

All countries that are not members of the OECD.

About this review

For 65 years, the *BP Statistical Review of World Energy* has provided high-quality objective and globally consistent data on world energy markets. The review is one of the most widely respected and authoritative publications in the field of energy economics, used for reference by the media, academia, world governments and energy companies. A new edition is published every June.

Online tools and resources

Key information

All the tables and charts found in the latest printed edition are available at bp.com/statisticalreview plus a number of extras, including:

- Historical data from 1965 for many sections.
- Additional data for refined oil production demand, natural gas, coal, hydroelectricity, nuclear energy, electricity, renewables and CO₂ emissions.
- PDF versions and PowerPoint slide packs of the charts, maps and graphs, plus an Excel workbook of the data.
- Regional and country factsheets.
- Videos and speeches.

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You can view predetermined reports or chart specific data according to energy type, region, country and year.
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 **Energy Outlook**
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Group chief executive's introduction

Energy in 2015 – slow demand growth amid plentiful supply.



Welcome to the *BP Statistical Review of World Energy*. This is the 65th edition of the *Statistical Review*, an important milestone for a publication that has traced developments in global energy markets since 1951, a year when coal provided more than half of the world's energy and the price of oil was around \$16 (in today's money). Remarkably, the share of oil in global energy then was almost identical to its share today, at a little over 30%.

We believe the *BP Statistical Review* contributes to the world's understanding of energy markets by providing timely and objective data to help inform discussions, debates and decision-making. Its annual data helps us better interpret the swings and fluctuations that we are living through, and the historical data provide an important context for gauging where we may be heading next.

This year's edition records the data for 2015, a year in which significant long-term trends in both the global demand and supply of energy came to the fore.

On the demand side, we are seeing a gradual deceleration in global energy consumption as the huge boost from globalization and Chinese industrialization slowly subsides. That slowing was compounded last year by continuing weakness in the global economy. As a result, global primary energy consumption grew by just 1.0% in 2015, similar to the rate of growth seen in 2014, but much slower than the average seen over the past decade. Much of this weakness was driven by China, where energy consumption grew at its slowest rate in almost 20 years. Even so, China remained the world's largest growth market for energy for a fifteenth consecutive year.

The supply of energy in recent years has been driven by different factors, notably technological advances that have increased the range and availability of different fuels. The US shale revolution has unlocked huge swathes of oil and gas resources. And rapid technological gains have supported strong growth in renewable energy, led by wind and solar power. These advances meant that, despite the weakness of energy demand, oil, natural gas and renewable energy all recorded solid growth in 2015. Their gain was at the expense of coal, which saw its largest fall on record, taking its share within primary energy to its lowest level since 2005.

As we know, the contrasting trends in the demand and supply of energy had major effects on energy prices, with oil, gas and coal prices all falling sharply last year. These price declines played a key role in prompting adjustments in energy markets: boosting demand in some markets, most notably oil; curtailing supply and

shifting the fuel mix in others. The extent of this adjustment bodes well for the future stability of our industry.

The combination of slow demand growth and a shift in the fuel mix away from coal towards natural gas and renewable energy had important implications for carbon emissions. In particular, carbon emissions from energy consumption are estimated to have been essentially flat in 2015, the lowest growth in emissions in nearly a quarter of a century, other than in the immediate aftermath of the financial crisis. Last year was of course significant for the UN-led COP21 meetings in Paris and the historic agreement to tackle climate change. BP supports those aims and is committed to playing its part in helping to achieve them. The stalling in the growth of carbon emissions in 2015 is a step in the right direction. But it is only a small step: the scale of the challenge remains substantial, requiring major shifts in both energy efficiency and the fuel mix.

Our industry is living through a period of profound change. But that is nothing new: the past 65 years have seen huge changes to the global energy landscape. Our task as an industry is to take the steps necessary to provide the energy to meet the world's growing demand and ensure our sector remains resilient to the many factors that may buffet us in the near term. We must continue to invest in energy, in all its forms, to meet future needs. That is no easy task and requires fine judgements – judgements that can be more confidently made when based on the kind of solid data and analysis provided by the *Statistical Review*. The need for BP's *Statistical Review* over the next 65 years is likely to be just as great as in the past.

Let me conclude by thanking BP's economics team and all those who helped us prepare this review – in particular those in the governments of many countries around the world who have contributed their official data again this year. Thank you for your continuing cooperation and transparency.



Bob Dudley
Group chief executive
June 2016

Welcome to the BP Statistical Review of World Energy.

On the demand side, we are seeing a **gradual deceleration in global energy consumption as the huge boost from globalization and Chinese industrialization slowly subsides**. That slowing was compounded last year by continuing weakness in the global economy. As a result, **global primary energy consumption grew by just 1.0% in 2015**, similar to the rate of growth seen in 2014, **but much slower than the average seen over the past decade**. Much of this weakness was driven by China, where energy consumption grew at its slowest rate in almost 20 years. Even so, **China remained the world's largest growth market for energy for a fifteenth consecutive year**.

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Growth in global primary energy consumption remained low in 2015; and the fuel mix shifted away from coal towards lower-carbon fuels.



▲ The six-level Puxi Viaduct in Shanghai is one of city's busiest interchanges and is used by thousands of vehicles every hour.

▼ In the UK the National Grid supplies electricity using about 7,200km of power cable line and 690km of underground cable.

+1.0%

Growth of global primary energy consumption, well below the 10-year average of 1.9%.

+1.5%

Growth of Chinese primary energy consumption, the world's largest increment.



Global primary energy consumption increased by just 1.0% in 2015, similar to the below-average growth recorded in 2014 (+1.1%) and well below its 10-year average of 1.9%. Other than the recession of 2009, this represented the lowest global growth since 1998.

Consumption growth was below the 10-year average for all regions except Europe & Eurasia; emerging economies accounted for 97% of the increase in global consumption. OECD consumption experienced a small increase, with growth in Europe offsetting declines in the US and Japan. Chinese consumption slowed further, but still recorded the world's largest increment in primary energy consumption for the fifteenth consecutive year. **Russia recorded the largest volumetric decline in primary energy consumption.**

By fuel, only oil and nuclear power grew at above-average rates, with oil gaining global market share for the first time since 1999. Renewables in power generation continued to grow robustly, to nearly 3% of global primary energy consumption, while coal consumption recorded the largest percentage decline on record. **Global CO₂ emissions from energy are estimated to have been essentially flat.**

Prices for all fossil fuels fell in 2015 for all regions. **Crude oil prices recorded the largest decline on record in dollar terms, and the largest percentage decline since 1986.** The annual average price for Brent, the international crude oil benchmark, declined by 47%, reflecting a growing imbalance between global production and consumption. The differential between Brent and the US benchmark West Texas Intermediate (WTI) narrowed to its smallest level since 2010.

Natural gas prices fell in all regions, with the largest percentage declines in North America; the US benchmark Henry Hub fell to its lowest level since 1999. Coal prices around the world fell for the fourth consecutive year.

Energy developments

Oil remained the world's leading fuel, accounting for 32.9% of global energy consumption. Although emerging economies continued to dominate the growth in global energy consumption, growth in these countries (+1.6%) was well below its 10-year average of 3.8%.

Emerging economies now account for 58.1% of global energy consumption. Chinese consumption growth slowed to just 1.5%, while **India (+5.2%) recorded** another robust increase in consumption. OECD consumption increased slightly (+0.1%), compared with an average annual decline of 0.3% over the past decade.

A rare increase in EU consumption (+1.6%) more than offset declines in the US (-0.9%) and Japan (-1.2%), where consumption fell to the lowest level since 1991.



Oil's share of global energy consumption, the first increase since 1999.



Prices

Dated Brent averaged \$52.39 per barrel in 2015, a decline of \$46.56 per barrel from the 2014 level and the lowest annual average since 2004. Crude oil prices rose in early 2015 as global consumption rebounded and US production began to register month-on-month declines. But strong growth in OPEC production, particularly in Iraq and Saudi Arabia, caused prices to fall sharply later in the year. The Brent – WTI differential narrowed for a third consecutive year, to \$3.68 per barrel.

Consumption and production

Global oil consumption grew by 1.9 million barrels per day (b/d), or 1.9% – nearly double the recent historical average (+1%) and significantly stronger than the increase of 1.1 million b/d seen in 2014. The relative strength of consumption was driven by the OECD countries, where consumption increased by 510,000 b/d (+1.1%), compared with an average decline of 1.1% over the past decade. Growth was well above recent historical averages in the **US (+1.6%, or 290,000 b/d)** and the **EU (+1.5%, or 200,000 b/d)**, while Japan (-3.9%, or -160,000 b/d) recorded the largest decline in oil consumption. Outside of the OECD, net oil importing countries recorded significant increases: China (+6.3%, or +770,000 b/d) once again accounted for the largest increment to demand, **while India (+8.1%, or 310,000 b/d) surpassed Japan as the world's third-largest oil consumer.** But this was offset by slower growth in oil producers, such that oil demand growth in the non-OECD as a whole (+2.6%, or 1.4 million b/d) was below its recent historical average.

Global oil production increased even more rapidly than consumption for a second consecutive year, rising by 2.8 million b/d or 3.2%, the strongest growth since 2004. **Production in Iraq (+750,000 b/d) and Saudi Arabia (+510,000 b/d) rose to record levels**, driving an increase in OPEC production of 1.6 million b/d to 38.2 million b/d, exceeding the previous record reached in 2012.

Production outside OPEC slowed from last year's record growth but still grew by 1.3 million b/d. **The US (+1 million b/d) had the world's largest annual growth increment and remained the world's largest oil producer.** Elsewhere, production growth in Brazil (+180,000 b/d), Russia (+140,000 b/d), the UK and Canada (+110,000 b/d each) was offset by declines in Mexico (-200,000 b/d, the world's largest decline), Yemen (-100,000 b/d) and elsewhere.

Refining and trade

Global crude runs rose by 1.8 million b/d (+2.3%), more than triple their 10-year average growth, despite declines in South & Central America, Africa and Russia. Strong refining margins lifted crude runs by 1 million b/d in the OECD, with growth in Europe (+740,000 b/d) the highest since 1986. In contrast, global refining capacity grew by only 450,000 b/d, the smallest increase in 23 years. Delayed expansion in China, combined with closures in Taiwan and Australia, resulted in a fall in Asian capacity for the first time since 1988. Global refinery utilization rose by 1 percentage point to 82.1%, the fastest increase in five years.

After barely growing in 2014, global trade of crude oil and refined products expanded by 3 million b/d (+5.2%) last year, the largest increase since 1993. Crude oil trade was lifted by growing exports from the Middle East (+550,000 b/d), while Europe and China accounted for the largest increases in imports (+770,000 b/d and +530,000 b/d respectively). Growth in refined product exports was again led by the US (+470,000 b/d); the country's net oil imports fell to 4.8 million b/d, the lowest since 1985.

🔥 Natural gas

🚂 Coal

+5.4%

Growth of US gas production, the world's largest increment.

-1.8%

Decline in global coal consumption, the largest on record.

▼ A coal digger extracting coal at a mine.

▲ A gas rig at BP's Alice well site near Durango, Colorado.



Gas

Consumption and production

World natural gas consumption grew by 1.7% in 2015, a significant increase from the very weak growth (+0.6%) seen in 2014 but still below the 10-year average of 2.3%. As with oil, consumption growth was below average outside the OECD (+1.9%, accounting for 53.5% of global consumption) but above average in the OECD countries (+1.5%). Among emerging economies, **Iran (+6.2%) and China (+4.7%) recorded the largest increments to consumption**, even though growth in China was sluggish compared with a 10-year average growth of 15.1%. Meanwhile, **Russia (-5%) recorded the largest volumetric decline, followed by the Ukraine (-21.8%).** Among OECD countries, the US (+3%) accounted for the largest growth increment, while EU consumption (+4.6%) rebounded after a large decline in 2014. **Globally, natural gas accounted for 23.8% of primary energy consumption.**

Global natural gas production grew by 2.2%, more rapidly than consumption but below its 10-year average of 2.4%. As with consumption, the US (+5.4%) recorded the largest growth increment, with Iran (+5.7%) and Norway (+7.7%) also recording significant increases in production. Growth was above average in North America, Africa, and Asia Pacific. EU production once again fell sharply (-8%), with the Netherlands (-22.8%) recording the world's largest decline. Large volumetric declines were also seen in Russia (-1.5%) and Yemen (-71.5%).

Trade

Global natural gas trade rebounded in 2015, rising by 3.3%. **Pipeline shipments increased by 4%, driven by growth in net pipeline exports from Russia (+7.7%) and Norway (+7%).** The largest volumetric increases in net pipeline imports were in Mexico (+44.9%) and France (+28.8%). Global LNG trade increased by 1.8%. Export growth was led by Australia (+25.3%) and Papua New Guinea (+104.8%), offsetting declines in shipments from Yemen (-77.2%). Higher net LNG imports for Europe (+15.9%) and rising Middle Eastern imports (+93.8%) were partly offset by declines in net imports in South Korea (-10.4%) and Japan (-4%). International natural gas trade accounted for 30.1% of global consumption; the pipeline share of global gas trade rose to 67.5%.

Coal

Global coal consumption fell by 1.8% in 2015, well below the 10-year average annual growth of 2.1% and the largest percentage (and volumetric) decline in our data set. All of the net decline was accounted for by the US (-12.7%, the world's largest volumetric decline) and China (-1.5%), **partially offset by modest increases in India (+4.8%) and Indonesia (+15%).** Global coal production fell by 4%, with large declines in the US (-10.4%), Indonesia (-14.4%), and China (-2%). **Coal's share of global primary energy consumption fell to 29.2%, the lowest share since 2005.**



Other fuels

+213 terawatt-hours
Growth in renewable
power generation, the largest
increment on record.

+0.1%
Increase in global CO₂
emissions from fossil
fuel use.



▲ Sugar cane reception, preparation and juice extraction facilities at BP Biofuels' Ituiutaba plant in Brazil.

▼ Wind turbines at the Goshen wind farm in Idaho Falls, Idaho.



Nuclear and hydroelectric

Global nuclear output grew by 1.3%, with China (+28.9%) accounting for virtually all of the increase. China has passed South Korea to become the fourth-largest supplier of nuclear power. Elsewhere, increases in Russia (+8%) and South Korea (+5.3%) offset declines in Sweden (-12.6%) and Belgium (-22.6%). EU output (-2.2%) fell to the lowest level since 1992. **Nuclear power accounted for 4.4% of global primary energy consumption.**

Global hydroelectric output grew by a below average 1%, compared with a 10-year average of 3%. China (+5%) remains by far the world's largest producer of hydroelectricity; as with nuclear power, China accounted for all of the net global increase, even though growth in percentage terms was less than half the recent historical average. Elsewhere, growth in Turkey (+64.6%, following a very weak 2014) and Scandinavia was offset by drought conditions in Italy, Spain and Portugal (-28.6% combined) and Brazil (-3.3%). Hydroelectric output accounted for 6.8% of global primary energy consumption.

Renewables

Renewable energy sources in power generation continued to increase in 2015, reaching **2.8% of global energy consumption**, up from 0.8% a decade ago.

Renewable energy used in power generation grew by 15.2%, slightly below the 10-year average growth of 15.9% but a record increment (+213 terawatt-hours), which was roughly equal to all of the increase in global power generation. Renewables accounted for 6.7% of global power generation. China (+20.9%) and Germany (+23.5%) recorded the largest increments in renewables in power generation. Globally, wind energy (+17.4%) remains the largest source of renewable electricity (52.2% of renewable generation), with Germany (+53.4%) recording the largest growth increment. Solar power generation grew by 32.6%, with China (+69.7%), the US (+41.8%) and Japan (+58.6%) accounting for the largest increases

China overtook Germany and the US to become the world's top generator of solar energy. Global biofuels production grew by just 0.9%, well below the 10-year average of 14.3%. Brazil (+6.8%) and the US (+2.9%) accounted for essentially all of the net increase, partly offset by large declines in Indonesia (-46.9%) and Argentina (-23.9%).

Carbon dioxide emissions **Emissions of CO₂ from energy consumption** **increased by only 0.1% in 2015.** Other than the recession of 2009, this represented the lowest growth rate since 1992. This encouraging outcome was driven by a combination of slightly slower energy consumption growth and a shift in the global fuel mix (with lower global coal consumption partly offset by growth in natural gas and non-fossil fuels). Emissions growth was below average in every region except Europe & Eurasia. **The US (-2.6%) and Russia (-4.2%) accounted for the largest absolute declines in emissions, while India (+5.3%) saw the largest increase.** Chinese emissions declined for the first time since 1998. This year's estimates on CO₂ emissions reflect a more detailed breakdown of fuel consumption by product type, as well as allowances for non-combusted hydrocarbons. See *bp.com/statisticalreview* for a detailed explanation of the revised methodology.